

Titration of the *Escherichia coli* DnaA protein to excess *datA* sites causes destabilization of replication forks, delayed replication initiation and delayed cell division

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Summary

In *Escherichia coli*, the level of the initiator protein DnaA is limiting for initiation of replication at *oriC*. A high-affinity binding site for DnaA, *datA*, plays an important role. Here, the effect of extra *datA* sites was studied. A moderate increase in *datA* dosage (~ fourfold) delayed initiation of replication and cell division, but increased the rate of replication fork movement about twofold. At a further increase in the *datA* gene dosage, the SOS response was induced, and incomplete rounds of chromosome replication were detected. Overexpression of DnaA protein suppressed the SOS response and restored normal replication timing and rate of fork movement. In the presence of extra *datA* sites, cells showed a dependency on PriA and RecA proteins, indicating instability of the replication fork. The results suggest that wild-type replication fork progression normally includes controlled pausing, and that this is a prerequisite for normal replication fork function.

Introduction

Initiation of chromosome replication in *Escherichia coli* is a complex process involving several different proteins, the main actor, however, being the DnaA protein (Kornberg and Baker, 1992). DnaA binds to five recognition sites called DnaA boxes within *oriC* (Fuller *et al.*, 1984; Schaper and Messer, 1995), unwinds the AT-rich region adjacent to the DnaA boxes (Bramhill and Kornberg, 1988) and forms the open complex, aided by HU and/or IHF and transcriptional activation (Skarstad *et al.*, 1990; Hwang

and Kornberg, 1992). The unwound region provides an entry site for the DnaB/DnaC helicase/helicase loader, followed by DnaG primase, which synthesizes RNA primers. Finally, the DNA polymerase III holoenzyme is loaded to synthesize new DNA. DNA gyrase and SSB are required to remove the supercoils and stabilize single-stranded (ss)DNA regions respectively (Kornberg and Baker, 1992).

Each origin is initiated only once per cell cycle. Reinitiation of newly initiated origins is prevented by the following mechanisms (Boye *et al.*, 2000): (i) sequestration of *oriC* (Campbell and Kleckner, 1990; Lu *et al.*, 1994); (ii) regulation of the activity of DnaA (Katayama *et al.*, 1998; Speck *et al.*, 1999; Kato and Katayama, 2001); and (iii) reduction in the level of free DnaA (Kitagawa *et al.*, 1998). The SeqA protein is involved in the sequestration of newly replicated hemimethylated DNA (von Freiesleben *et al.*, 1994; Lu *et al.*, 1994) and apparently prevents the formation of an initiation complex during the sequestration period (Boye *et al.*, 1996; Wold *et al.*, 1998; Skarstad *et al.*, 2000; Taghbalout *et al.*, 2000). A reduction in the amount of active DnaA protein (ATP-DnaA) occurs by the combined activity of the replicative polymerase, the β -clamp and the Hda protein (Katayama *et al.*, 1998; Kato and Katayama, 2001). A reduction in the amount of free DnaA protein during the origin inactivation is caused by both sequestration of the *dnaA* promoter, which prevents transcription of the *dnaA* gene (Campbell and Kleckner, 1990; Bogan and Helmstetter, 1997), and titration of free DnaA protein by newly replicated DnaA boxes (Hansen *et al.*, 1991; Løbner-Olesen *et al.*, 1994; Kitagawa *et al.*, 1998).

In the *E. coli* chromosome, there are 308 DnaA boxes (TTAT(C/A)CA(C/A)A) that have variable affinity to DnaA (Schaper and Messer, 1995). Among these DnaA boxes, a strong binding region, *datA* (DnaA titration) harbouring five boxes, has been identified (Kitagawa *et al.*, 1996). The *datA* site titrates unusually large amounts of DnaA protein *in vivo* (Kitagawa *et al.*, 1996) and seems to be the main contributor to the DnaA protein titration mechanism (Ogawa *et al.*, 2002). It has been suggested that the DnaA molecules bound at *datA* act as a reservoir of DnaA, thus affecting the timing of replication initiation (Kitagawa *et al.*, 1998). Lately, it has been found that cells with high levels of *datA* completely shut off initiation of replication from *oriC* (Morigen *et al.*, 2001).

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Table 1. Average cell mass, DNA content, *oriC:terC* ratio and growth rate in strain MG1655 harbouring *datA* plasmids.

Plasmid	Doubling time (min)	Plasmid ^a borne <i>datA</i>	Dosage of <i>datA</i>	Relative ^b cell mass	Relative ^b DNA content per cell	Relative ^b DNA per mass	<i>oriC/terC</i> ^c
No plasmid	35	0	3.5 ^d	1	1	1	3.1
MiniR1- <i>datA</i>	39	10	14 ^e	1.6	1.2	0.8	1.8
pACYC177- <i>datA</i> 91	91	40	44 ^e	3.5	1.6	0.5	1.4

a. Copy numbers of *datA* plasmids per *terC* were determined by Southern hybridization in exponentially growing cells.

b. The values are relative to the wild type.

c. Determined by Southern hybridization in exponentially growing cells.

d. As *datA* is located close to *oriC*, *datA* dosage on the chromosome was estimated by calculation of origins per cell from data of flow cytometry of cells treated with rifampicin and cephalixin.

e. Total dosage of *datA* on both chromosomes and plasmids.

The values are the average of three individual experiments.

In this work, we studied the effects of several different *datA* gene dosages in chromosome replication and cell division.

Results

Wild-type MG1655 cannot be transformed with high-copy-number plasmids carrying datA sites

To monitor the effects of different levels of *datA* on regulation of chromosome replication initiation from *oriC*, we used a series of plasmids with different copy numbers, all carrying the *datA* locus: pMOR1 (MiniR1-*datA*), pMOR6 (pACYC177-*datA*), pMOR5 (pBR322-*datA*), pMOR2 (pNEB193-*datA*) and corresponding vectors as control. A wild-type strain (MG1655; Guyer *et al.*, 1981; Jensen, 1993) was transformed with the plasmids, and viability of the transformants was measured by restreaking the transformants on the same kind of selective plates. Transformation with plasmids of lower copy numbers (MiniR1-*datA*, pACYC177-*datA*) yielded transformants with full viability after restreaking. Transformation with plasmids of higher copy numbers (pBR322-*datA*, pNEB193-*datA*) resulted in transformed cells capable of forming colonies initially, but none was viable after restreaking. All transformations with control vectors yielded viable transformants. The doubling times of the MiniR1-*datA*- and pACYC177-*datA*-carrying strains were 39 and 91 min, respectively, compared with 35 min in the wild type (Table 1). The results show that the presence of high levels of *datA* inhibits cell growth. This is in accordance with previous reports in which titration of DnaA protein by excess *datA* sites reduced or completely shut down initiation of replication from *oriC* (Kitagawa *et al.*, 1998; Morigen *et al.*, 2001).

Initiation of replication and cell division are delayed in the presence of excess datA

To investigate further the inhibitory effect of *datA*, we measured cell mass, DNA content per cell and origin-to-

terminus ratio in the above-mentioned strains by flow cytometry and Southern blotting (see *Experimental procedures*). Introduction of increasing numbers of *datA* loci led to a decrease in the DNA concentration by 20% at ≈fourfold *datA* dosage (MiniR1-*datA*) and 50% at ≈12-fold *datA* (pACYC177-*datA*) (Table 1). The cell mass was increased 1.6 (MiniR1-*datA*) and 3.5 times (pACYC177-*datA*) relative to the wild type (Table 1). The *oriC* to *terC* ratio was reduced by about 40% (MiniR1-*datA*) and 55% (pACYC177-*datA*) relative to the wild type (Table 1). Thus, in the presence of excess *datA*, cells became larger and had a lower DNA concentration and origin:terminus ratio than the control.

The origin content of individual cells was investigated by flow cytometry of samples treated with rifampicin and cephalixin (Fig. 1). In wild-type cells with control vectors, initiation occurred at two origins early in the cell cycle. Thus, most cells in the two cultures contained four origins (Fig. 1A and C). Young cells that had not yet initiated contained two. The largest cells contained eight origins and had initiated replication before division in the previous cycle. The fact that cultures contain both two-origin and eight-origin cells results from cell-to-cell variability in the exact time of division relative to the time of initiation.

Most cells with fourfold more *datA* contained two origins; the fraction of cells with two origins was almost twice as high as in the control (Fig. 1B). This indicates that initiation occurred later in the division cycle in these cells because of an inhibitory effect of extra *datA* on initiation from *oriC*. Cells harbouring 12-fold *datA* did not yield histograms with separate peaks after rifampicin and cephalixin treatment (Fig. 1D), indicating either lack of replication elongation or partial degradation of the DNA. The presence of 12-fold *datA* caused induction of the SOS response (see below).

In summary, a lack of free DnaA protein, obtained by the presence of fourfold excess *datA* sites, led to (i) delayed initiation of replication relative to the division cycle and relative to cell mass; and (ii) delayed cell division

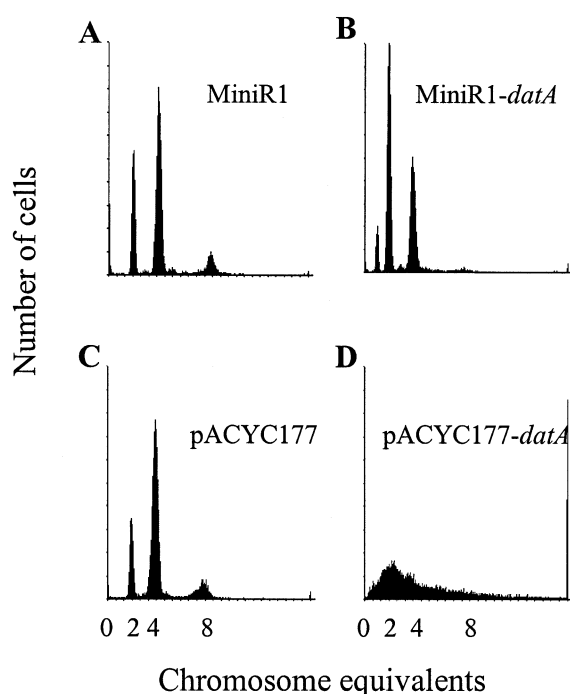


Fig. 1. Excess *datA* delays initiation of replication and affects the cells' ability to complete rounds of replication. Strain MG1655 containing the indicated plasmids was grown in ABTG-CAA medium and treated with rifampicin and cephalaxin for four or five generations. Samples were measured by flow cytometry. Distinct peaks represent the accumulation of cells with integral numbers of chromosomes that reflect the numbers of origins at the time of drug action. A total of 10 000 cells is shown in each panel at the same scale.

relative to cell mass, which resulted in a 60% increase in cell mass.

To investigate whether the nucleoid distribution was affected by excess *datA*, cells briefly treated with chloramphenicol to condense the nucleoids were studied by microscopy (Begg and Donachie, 1991). In cultures of cells with fourfold *datA*, 18% of the cells contained two separated nucleoids, compared with 7% in the control (Table 2 and Fig. 2). This increase in the fraction of cells containing separated nucleoids shows that cells with excess *datA* have no problems with segregation. On the contrary, these cells seem to go through the cell cycle with an extended interval of time between completion of segregation and cell division. Thus, the delay in cell division caused by the presence of excess *datA* was not caused by problems with segregation. Instead, cell division must

be affected more directly by the delay in initiation and the lack of free DnaA protein.

Increased DnaA availability suppresses effects of the excess datA sites

In order to see whether an increased DnaA concentration suppresses the effects of extra *datA* sites, we first introduced a DnaA-overproducing plasmid, pFH871, into the wild-type strain. DnaA protein was overexpressed $\approx$ eightfold (data not shown). Subsequently, this strain was transformed with the plasmids carrying the *datA* locus. In contrast to the wild-type strain, the DnaA-overproducing strain could be transformed with the high-copy-number plasmid pBR322-*datA*, which yields 34-fold excess *datA*. The DNA concentrations in the DnaA-overproducing cells carrying MiniR1-*datA* ($\approx$ fourfold *datA*) and pSC101-*datA* ($\approx$ sevenfold *datA*) were about the same as for the wild type (Table 3). Also, the cell mass and DNA content per cell of the former strain were essentially normal, whereas for the latter strain, both parameters were increased (Table 3).

In the presence of excess DnaA (and no plasmid-borne *datA*), about half the cells contained four origins, and the rest contained five, six, seven or eight origins (Fig. 3B). This origin distribution indicates that initiation no longer occurred in two-origin cells, but in four-origin cells (i.e. 'earlier'), and that initiations were not synchronous at the four origins. The result confirms that overinitiation occurs when excess DnaA is provided (Skarstad *et al.*, 1989;

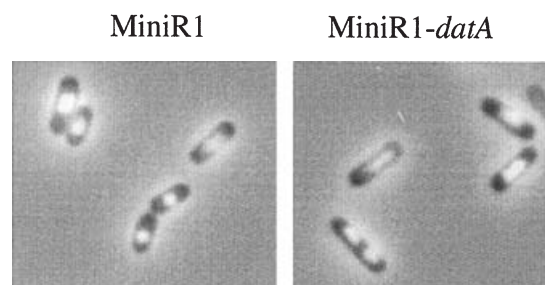


Fig. 2. Cell division, but not chromosome segregation, is delayed in the presence of the excess *datA*. Exponentially growing MG1655 carrying MiniR1-*datA* or MiniR1 in ABTG-CAA medium were treated for 15 min with $300 \mu\text{g ml}^{-1}$ chloramphenicol and $10 \mu\text{g ml}^{-1}$ cephalaxin to condense the nucleoids and inhibit cell division during condensation. Samples were stained with Hoechst and viewed by phase-contrast and fluorescence microscopy.

Table 2. Nucleoid distribution^a and growth rate in strain MG1655 with and without *datA* plasmid.

Plasmid	Doubling time (min)	One-nucleoid cells (%)	Two-nucleoid cells (%)	Uncountable cells (%)
MiniR1	35 $\pm$ 0.3	83 $\pm$ 1.0	7 $\pm$ 0.3	10 $\pm$ 0.3
MiniR1- <i>datA</i>	40 $\pm$ 0.6	71 $\pm$ 1.5	18 $\pm$ 0.9	11 $\pm$ 0.9

a. More than 280 cells for each experiment were included in the nucleoid counting. The values are the average of three experiments.

Table 3. Average cell mass, DNA content, *oriC:terC* ratio and growth rate in strain MG1655 harbouring *datA* plasmids and producing excess DnaA protein.

Plasmid	Doubling time(min)	Plasmid ^a borne <i>datA</i>	Dosage of <i>datA</i>	Relative ^b cell mass	Relative ^b DNA content per cell	Relative ^b DNA per mass	<i>oriC/terC</i> ^c
No plasmid	35	0	3.5 ^d	1	1	1	3.1
pFH871 ^e	42	0	5 ^d	1.2	1.3	1.1	3.1
pFH871 + MiniR1- <i>datA</i>	38	7	12 ^f	1.2	1.2	1.0	2.0
pFH871 + pSC101- <i>datA</i>	40	20	25 ^f	2.1	1.8	0.9	1.9

a. Copy numbers of *datA* plasmids were determined by Southern hybridization in exponentially growing cells.

b. The values are relative to the wild type.

c. Determined by Southern hybridization in exponentially growing cells.

d. As *datA* is located close to *oriC*, *datA* dosage on the chromosome was estimated by calculation of origins per cell from data of flow cytometry of cells treated with rifampicin and cephalixin.

e. DnaA overproducer.

f. Total dosage of *datA* on both chromosome and plasmid.

The values are the average of three experiments.

Atlung and Hansen, 1993). Excess DnaA protein also led to a shift towards higher origin content in cells carrying a fourfold excess of *datA* (Fig. 3D). The cells with sevenfold excess *datA* and eightfold DnaA showed an almost wild-type distribution of origins (Fig. 3F), indicating that the two opposing effects balanced each other. Some asynchrony was, however, observed, possibly indicating that the amount of excess DnaA and number of extra *datA* sites did not exactly match up in all cells. Cells harbouring pSC101-*datA* without overproduction of DnaA (Fig. 3E) did not seem to have severe problems with run-out of replication as seen when pACYC177-*datA* was present (Fig. 1D), yet the origin content per cell was increased rather than decreased relative to the wild-type strain (compare Fig. 3E and 3A). This possibly resulted from inhibition of cell division by the SOS response (see below). The strain with 34-fold excess *datA* did not yield a histogram with separate peaks (Fig. 3G), suggesting that the effects of pBR322-*datA* on replication cannot be completely suppressed by pFH871. The reason for this is probably that insufficient amounts of extra DnaA are produced to suppress the effects of 34-fold excess *datA*.

In summary, the results show that the increased DnaA availability suppresses the negative effect of extra *datA* sites.

Absence of SeqA protein alleviates the effects of high datA dosage

The ability of SeqA-less cells to tolerate *datA* plasmids was investigated. Transformation of strain MG1655Δ*seqA* with MiniR1-*datA* and pACYC177-*datA* yielded transformants, 100% of which were viable after restreaking. Transformation with pBR322-*datA* gave 10% viable transformants after restreaking. As the wild-type strain transformed with pBR322-*datA* did not yield any viable transformants after restreaking, this shows that the Δ*seqA*

strain tolerates a higher copy of the *datA* region than the wild-type strain. Introduction of the plasmid with the highest copy number (pNEB193-*datA*) did not produce any viable transformants. A twofold increase in *datA* dosage (MiniR1-*datA*) in the Δ*seqA* strain resulted in an improved growth rate, i.e. more similar to its *seqA*⁺ counterpart, and a reduction in the origin per terminus ratio by about 50% compared with the Δ*seqA* strain (Table 4), suggesting that extra *datA* inhibits the overinitiation that occurs in the absence of SeqA protein (von Freiesleben *et al.*, 1994; Lu *et al.*, 1994; Boye *et al.*, 1996). This result shows that extra *datA* sites can, to a certain extent, compensate for the lack of sequestration. The decrease in growth rate seen in the wild type by harbouring pACYC177-*datA* (Table 1, line 3) was improved in the Δ*seqA* cells (Table 4, line 4), showing that the absence of SeqA protein alleviated the negative effect of ≈10-fold extra *datA* on growth rate.

DnaA protein concentrations

DnaA protein concentrations of all strains were determined by immunoblotting (see *Experimental procedures*). In wild-type cells carrying MiniR1-*datA* or pACYC177-*datA*, DnaA concentrations were increased ≈twofold relative to the wild type without plasmid (Table 5). In the DnaA-overexpressing strain (carrying pFH871), the DnaA concentration was increased ≈eightfold relative to the wild type and was further increased to about 11-fold with the introduction of pBR322-*datA* (Table 5). This indicates that 34-fold extra *datA* leads to derepression of the *dnaA* promoter so that more DnaA protein is synthesized even when approximately eight times extra DnaA is available in the cell. It should be noted that not only transcription from the chromosomal *dnaA* promoter but also transcription from the *dnaA* promoters on each pFH871 plasmid is repressed by DnaA protein.

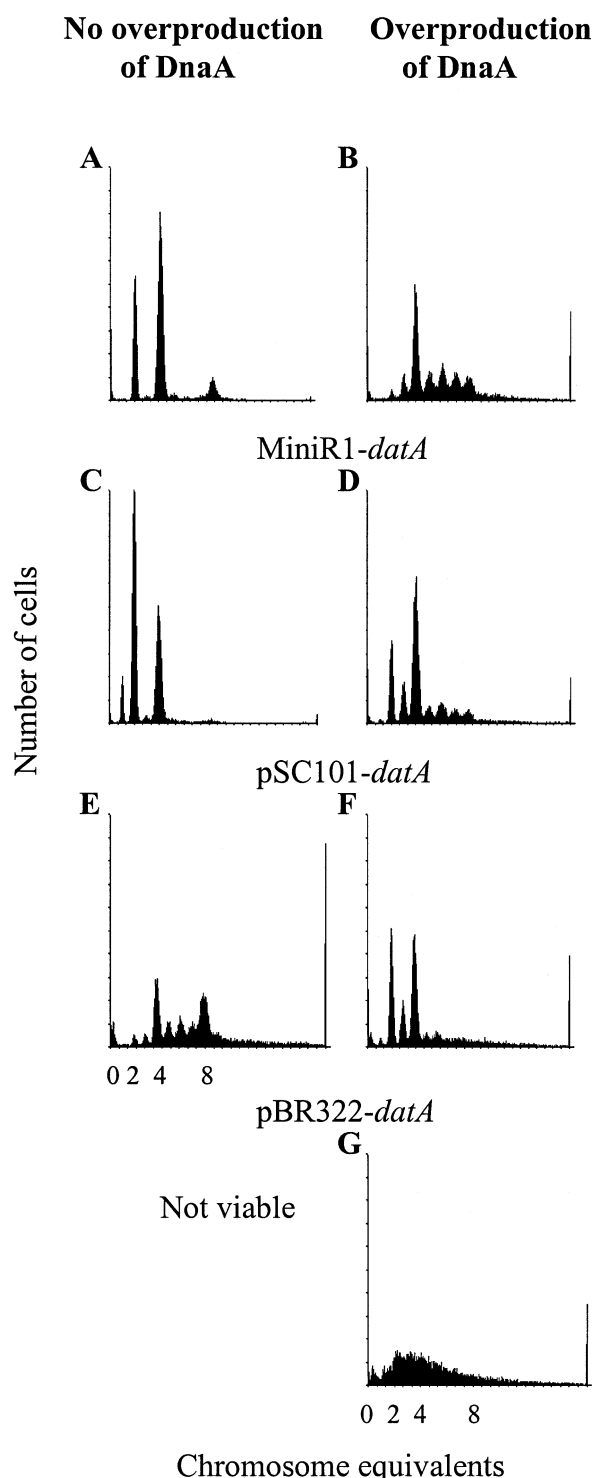


Fig. 3. Overexpression of DnaA protein suppresses the negative effects of the excess *dataA* sites. Strain MG1655 with (right) or without (left) plasmid pFH871 (DnaA overproducer), harbouring the indicated *dataA* plasmids, was grown in the ABTG-CAA medium and treated with rifampicin and cephalixin for four or five generations. A total of 10 000 cells measured by flow cytometry is shown in each panel at the same scale.

In the $\Delta seqA$ strain, the DnaA concentration was increased $\approx$ twofold relative to the wild type (Table 5), in accordance with results presented in previous reports (von Freiesleben *et al.*, 1994; 2000a; Torheim *et al.*, 2000). In the absence of SeqA protein, two different effects are apparent: (i) overinitiation of replication, which increases the *dnaA* gene dosage (von Freiesleben *et al.*, 1994; Lu *et al.*, 1994); and (ii) lack of sequestration of the *dnaA* promoter so that the *dnaA* gene may be transcribed throughout the cell cycle (Campbell and Kleckner, 1990; Bogan and Helmstetter, 1997). The DnaA concentration in the $\Delta seqA$ strain was, however, not increased further with the presence of excess *dataA* (Table 5). This apparent lack of response might be explained by the effects on gene dosage and transcription working in opposite directions, resulting in (i) a reduction in *dnaA* gene dosage because of inhibition of overinitiation; and (ii) increased transcription from each *dnaA* gene as a result of derepression of the promoters. Such a combination of a reduction plus an increase in total gene expression might explain the unchanged DnaA concentration observed.

Induction and reversal of the SOS response

The cells carrying around 10-fold extra *dataA* sites became larger than normal with a heterogeneous size distribution and some filamentation, which are features of the SOS response (Walker, 1984). To examine whether the excess *dataA* may induce the SOS response, we measured the activity of the SOS-controlled promoter of the *sfiA* gene fused to the *lacZ* reporter gene in the presence of increased *dataA* dosage. In this strain (ALO1186), we found a twofold or more increase in *lacZ* expression relative to the wild type with control vectors upon introduction of pSC101-*dataA* or pACYC177-*dataA* (Table 6, left column). This shows that about 10-fold extra *dataA* induces an SOS response. No significant increase in the expression of the reporter was observed in the presence of plasmid MiniR1-*dataA* (Table 6, left column), indicating that a fourfold excess of *dataA* does not induce the SOS response.

Expression of excess DnaA protein reverted the level of *lacZ* expression to that of the negative control (Table 6), showing that the *dataA*-induced SOS response can be suppressed by increased DnaA availability. The extra DnaA *per se* did not induce the SOS response (Table 6). The increased DnaA concentration provided from pFH871 was not sufficient to reduce *lacZ* expression to the level of the negative control in the presence of pBR322-*dataA* (Table 6). We conclude that a severe decrease in the free DnaA concentration induces the SOS response.

The *lexA3* mutation renders the LexA repressor non-cleavable and thus the SOS regulon non-inducible (Mount *et al.*, 1972). If the SOS response were important for cell

Table 4. Average cell mass, DNA content, *oriC:terC* ratio and growth rate in the $\Delta seqA$ strain harbouring *datA* plasmids.

Strain and plasmid	Doubling time (min)	Plasmid ^a borne <i>datA</i>	Dosage of <i>datA</i>	Relative ^b cell mass	Relative ^b DNA content per cell	Relative ^b DNA per mass	<i>oriC:terC</i> ^c
MG1655	35	0	3.5 ^d	1	1	1	3.1
MG1655 $\Delta seqA$							
No plasmid	46	0	5.3 ^d	1.8	1.8	1	5.2
MiniR1- <i>datA</i>	41	7	11 ^e	1.9	1.5	0.8	2.3
pACYC177- <i>datA</i>	65	47	51 ^e	3.2	1.4	0.4	2.9
pBR322- <i>datA</i> ^f	71	116	120 ^e	3.4	1.6	0.5	2.1

a. Copy numbers of *datA* plasmids were determined by Southern hybridization in exponentially growing cells.

b. The values are relative to the wild type.

c. Determined by Southern hybridization in exponentially growing cells.

d. As *datA* is located close to *oriC*, *datA* dosage on the chromosome was estimated by calculation of origins per cell from data of flow cytometry of cells treated with rifampicin and cephalixin.

e. Total dosage of *datA* on both chromosome and plasmid.

f. Not viable in the presence of SeqA.

The values are the average of three experiments.

survival, the *lexA3* mutant would be sensitive to the dosages of extra *datA* at which the SOS event was induced in the wild type. Any significant difference in transformation frequencies with different *datA* plasmids between the *lexA3* mutant and the control strain was not observed (data not shown). The colony size of the *lexA3* transformants was, however, much smaller than the control. The results suggest that the SOS event is not essential for cell survival in the presence of excess *datA*, but improves growth (see below).

The free DnaA concentration affects the replication fork movement

The above results show that titration of the DnaA protein by extra *datA* sites induces the SOS response. One way in which the SOS response may be induced is by the persistence of unpaired nucleotides in the 'lagging' DNA strand during the elongation phase of DNA synthesis (Walker, 1984; Sassanfar and Roberts, 1990). Therefore, we investigated whether lack of DnaA protein could lead to changes in replication elongation. To measure chain elongation rate, we compared changes in DNA histo-

grams of samples taken at different time intervals after addition of rifampicin and cephalixin (Fig. 4). The kinetics of this change reflects the rate of replication fork movement (Skarstad *et al.*, 1989; Skarstad and Wold, 1995). In histograms of the wild type (Fig. 4A), a peak representing cells with two chromosome equivalents appeared after 20 min and, after 40 min, this peak had become completely sharp. This indicates that replication in the two-chromosome cells was complete after 40 min. The four-chromosome peak and small eight-chromosome peak were completely sharp only after 90 min, indicating that replication of the chromosomes in the whole population was not complete until after 90 min. In histograms of cells harbouring fourfold extra *datA* (MiniR1-*datA*; Fig. 4B), two-, four- and eight-chromosome peaks were all sharp after 40 min, indicating that ongoing replication was finished after 40 min. Thus, the presence of fourfold extra *datA* significantly increased the replication rate relative to that of the wild type.

Table 5. DnaA protein concentration.

Plasmid	DnaA concentration (relative to wild type) in strains		
	MG1655	MG1655/pFH871 ^a	MG1655 $\Delta seqA$
No plasmid	1	7.7	2.2
MiniR1- <i>datA</i>	1.9	8.2	2.0
pACYC177- <i>datA</i>	2.2	— ^b	2.1
pBR322- <i>datA</i>	NV ^c	11.2	2.2

a. DnaA overproducer.

b. Not determined because of plasmid incompatibility.

c. Not viable.

The values are the average of three experiments.

Table 6. Induction and reversal of the SOS response.

Plasmid/treatment	Expression of <i>lacZ</i> (Miller units) in the strain	
	ALO1186 ^a	ALO1186/pFH871 ^b
pACYC177	33 ± 2.5	— ^c
pBR322	31 ± 1.8	36 ± 1.7
MiniR1- <i>datA</i>	34 ± 3.5	33 ± 1.5
pSC101- <i>datA</i>	69 ± 2.3	34 ± 1.2
pACYC177- <i>datA</i>	75 ± 0.3	— ^c
pBR322- <i>datA</i>	NV ^d	59 ± 1.8
pBR322 + UV ^e	222 ± 2.4	182 ± 1.5

a. ALO1186 carrying a SOS-controlled promoter, *sfia*, fused to the *lacZ* reporter gene.

b. DnaA overproducer.

c. Not determined because of plasmid incompatibility.

d. Not viable.

e. See *Experimental procedures*.

The values are the average of three experiments.

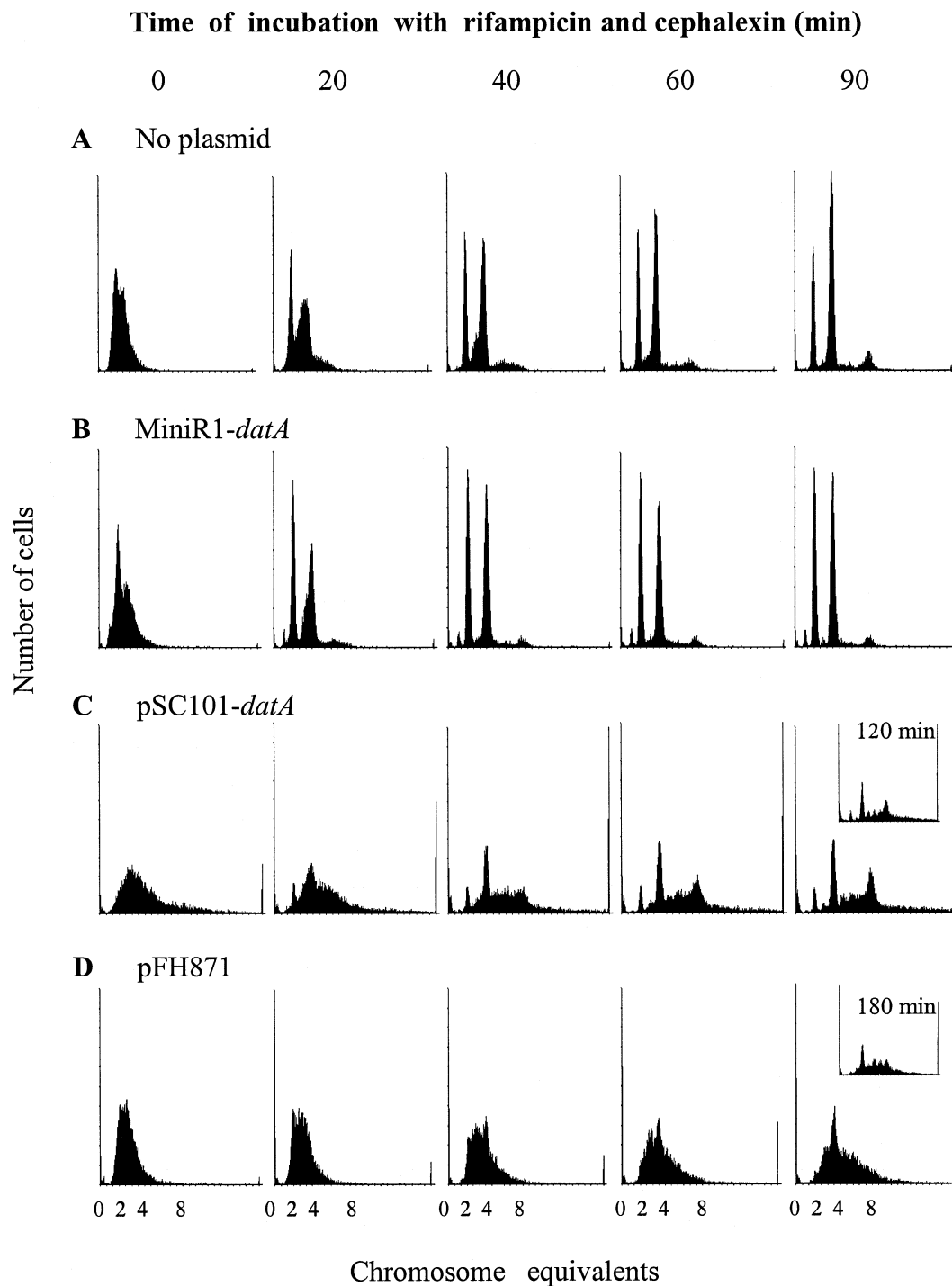


Fig. 4. The amount of free DnaA protein affects the rate of replication fork movement. Strain MG1655 without (A) and with MiniR1-*datA* (B), pSC101-*datA* (C), pFH871 (D) were grown in the ABTG-CAA medium and treated with rifampicin and cephalixin. Samples were taken at the indicated time intervals. A total of 10 000 cells measured by flow cytometry is shown in each panel at the same scale.

Further increase in the *datA* dosage to ≈ 12 -fold led, as mentioned above, to induction of the SOS response and to an inhibition of the ability to finish rounds of replication (Fig. 1D). Histograms of samples taken at different times

after addition of rifampicin and cephalixin showed that there was no net increase in DNA content (Fig. 5A). This may indicate a certain extent of DNA degradation as well as feeble DNA replication. Cells containing ≈ 10 -fold *datA*

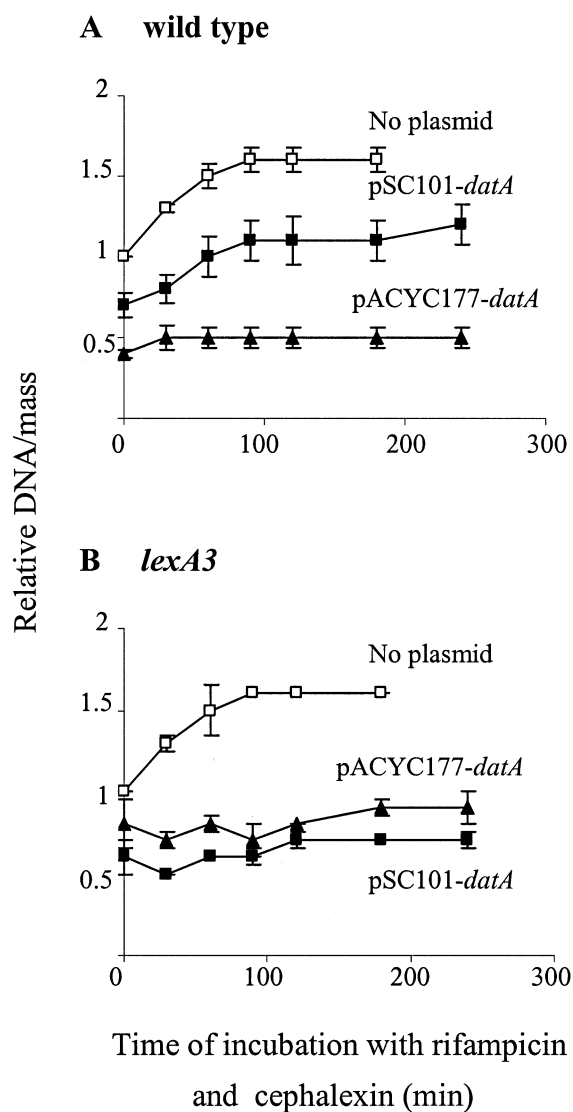


Fig. 5. Completion of rounds of replication after the addition of rifampicin in cells harbouring pSC101-*datA* is dependent on the SOS response. Wild-type (MG1655) (A) and *lexA3* mutant (B) cells harbouring the indicated plasmids were grown in the ABTG-CAA medium. Samples of rifampicin- and cephalixin-treated cells were taken at the indicated time intervals. The average DNA content per mass was calculated from flow cytometry histograms. The values are the average of three individual experiments, the standard error of the mean is indicated. DNA content/mass is relative to the wild type (A) or the *lexA3* mutant (B) without plasmid at 0 min after the addition of rifampicin and cephalixin.

(pSC101-*datA*) also showed induction of the SOS response and increased cell size, although the effect was not so severe (Table 6). DNA replication proceeded and histogram peaks became sharp, but not until after 120 min, showing that replication took more than 30 min longer than in the wild type (Figs 4C and 5A). This might indicate that some replication forks were arrested or paused. Also, a slight asynchrony phenotype was appar-

ent (Fig. 4C). It is possible that the SOS response may have induced alternative ways of replication, for instance iSDR (Magee *et al.*, 1992).

In *lexA3* mutant cells containing either pSC101-*datA* or pACYC177-*datA*, no net increase in DNA content could be detected after treatment with rifampicin and cephalixin (Fig. 5B). This result indicates that, in cells carrying pSC101-*datA*, induction of the SOS response was beneficial, whereas in cells carrying pACYC177-*datA*, the lack of DnaA was more extreme and caused a situation in which continuous replication could not be salvaged by the SOS response.

In cells overexpressing DnaA eightfold (Fig. 4D), four-, five-, six-, seven- and eight-chromosome peaks were not sharp until after 180 min. Thus, the replication rate in these cells was only half that of the wild type. Similar slow fork movement in the presence of excess DnaA has been reported in previous work (Skarstad *et al.*, 1989; Atlung and Hansen, 1993). These results suggest that excess free DnaA protein leads, directly or indirectly, to slower replication fork movement.

The priA, recA and recBC mutants are more sensitive to the extra datA than the wild type

Limiting amounts of free DnaA, caused by a fourfold excess of *datA*, led to an increased replication fork rate, i.e. a shortening of the C-period (Fig. 4). A further reduction in free DnaA, caused by a more than sevenfold excess of *datA*, had the effect of lengthening the C-period, and was accompanied by induction of the SOS response. From these findings, it is reasonable to suggest that the regulation and/or stability of replication forks is changed by excess *datA*, leading to a lack of necessary pauses at moderately limiting DnaA, and frequent breakdown of forks at severely limiting DnaA levels. The PriA protein is required for restarting arrested replication forks (Kogoma *et al.*, 1996; Michel *et al.*, 1997; Sandler and Marians, 2000). If replication fork stability were decreased in the presence of extra *datA*, the viability of a *priA* mutant with excess *datA* might be expected to be reduced. In wild-type cells, the transformation efficiency with MiniR1-*datA* was the same as that with control vector, but further increase in the *datA* dosage to ≈ 10 -fold by introduction of pSC101-*datA* reduced the transformation efficiency by 10-fold relative to the vector (Table 7). In the *priA* mutant, the transformation efficiencies with MiniR1-*datA* and pSC101-*datA* were 4- and 1000-fold lower than with the vectors, and 4- and 100-fold lower than that in the wild-type strain (Table 7). These results indicate an increased dependence on the PriA protein in the presence of excess *datA*. This suggests that optimally functioning replication forks are dependent on DnaA, either directly or indirectly.

Table 7. The *priA*, *recA* and *recBC* mutants are more sensitive to the extra *datA* than the wild type.

Incoming plasmid	Strain			
	Wild type	<i>priA</i> ^a	<i>recA56</i>	<i>recB21</i> , <i>recC22</i>
	Transformation efficiency ^b			
MiniR1	1.1×10^8	8.5×10^4	1.1×10^6	4.5×10^5
MiniR1- <i>datA</i>	1.0×10^8	2.4×10^4	3.8×10^5	4.5×10^5
pSC101	7.5×10^7	2.8×10^5	4.5×10^6	5.0×10^5
pSC101- <i>datA</i>	8.0×10^6	1.6×10^2	4.6×10^3	4.0×10^3

a. JJC1398 was cured of the PriA-producing plasmid immediately before preparation of competent cells.

b. Transformation efficiency was measured by formed colonies μg^{-1} plasmid DNA.

The results are from three independent experiments.

The RecBC proteins co-operate with the RecA protein to carry out recombinational repair of double-strand breaks and collapsed replication forks (Kowalczykowski *et al.*, 1994; Kuzminov, 1995a). *In vitro* data also suggest that RecA protein promotes the regression of stalled replication forks (Seigneur *et al.*, 2000; Robu *et al.*, 2001). The formation of RecA-dependent recombination intermediates is necessary for PriA/Pol II-dependent replication restart (Rangarajan *et al.*, 2002). As shown in Table 7, the transformation efficiencies were 1000- and 100-fold lower with pSC101-*datA* relative to the vector in *recA56* and *recB21 recC22* mutants.

These results indicate that cells harbouring 10-fold extra *datA* become dependent on recombination functions for replication. The reason for this may be that cells require extensive salvage of stalled replication forks. Alternatively, they require iSDR, which requires recombination proteins, to initiate replication. Cells harbouring only fourfold extra *datA* were also more sensitive than wild type to the loss of RecA. These cells replicate the chromosome faster than normal (Fig. 4) and may therefore be more prone to replication fork collapse and more dependent on salvage of broken replication forks.

Cells carrying excess *datA* are sensitive to UV light

If replication forks in cells carrying extra *datA* are compromised, it might be expected that these cells would be more sensitive than wild type to a DNA-damaging agent such as UV light. Electroporated wild-type cells, with or without the DnaA overproducer plasmid pFH871, were incubated with pSC101-*datA* or control vector, plated on selective plates and immediately exposed to UV light (Fig. 6). The transformation efficiency for pSC101-*datA* was 10^5 -fold lower than that for control vector after the cells were exposed to the UV light for 60 J m^{-2} , showing that the cells attempting to harbour *datA* plasmids were more sensitive to the effects of UV than the control. Over-

production of DnaA reversed the decreased transformation efficiency (Fig. 6). This shows that this effect caused by extra *datA* could also be suppressed by providing excess DnaA. Transformation efficiency of cells containing excess DnaA was the same as that of the control (Fig. 6). These results support our suggestion that the lack of free DnaA leads to problems with replication fork progression.

Discussion

Titration of DnaA by the datA region delays initiation of replication and cell division

In this work, we have investigated the effects of increasing numbers of *datA* sites on chromosome replication and cell growth. The results confirm earlier discoveries (Kitagawa *et al.*, 1996; 1998), indicating that titration by *datA* of DnaA contributes to determining at which point in the cell cycle initiation occurs. A fourfold increase in *datA* dosage led to a considerable delay in initiation of replication and a 60% increase in average cell mass. The cell mass at the time of initiation was not measured directly but, from the fact that initiation was delayed, can be taken to be more than 60% increased. Thus, an almost twofold increased initiation mass together with a twofold increased DnaA concentration allowed sufficient amounts of DnaA protein for initiation in the presence of fourfold *datA*. The increased

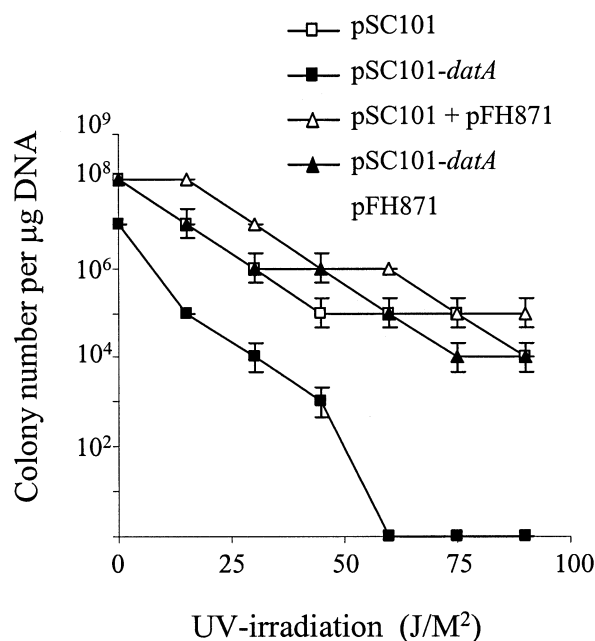


Fig. 6. Cells with extra *datA* are sensitive to UV light. Wild-type MG1655 was transformed with the indicated plasmids and plated on selective plates. The plated cells were irradiated by UV light (see *Experimental procedures*) at the doses indicated. The values are the average of three individual experiments, the standard error of the mean is indicated. pFH871 is a DnaA overproducer.

cell mass was caused by a delay in cell division and not by problems with segregation. It is not known whether this inhibition of cell division is caused directly by the lack of free DnaA protein, or whether the mechanism of inhibition is more indirect, for instance through the regulation of other gene products affecting division.

The DnaA protein has been shown to act as a transcription factor at the promoters of several different genes (reviewed by Messer and Weigel, 1997). This is in keeping with the large number of 9-mer *dnaA* boxes found throughout the chromosome and indicates that changes in DnaA protein availability and activity will change the expression of multiple genes simultaneously. DnaA binding sites have been found within the *ftsQ* and *ftsA* genes (Masters *et al.*, 1989; Smith *et al.*, 1996). Such sites as well as the more recently discovered 6-mer ATP-DnaA binding sites (Speck and Messer, 2001) may be important in coupling DnaA activity to the cell cycle and preventing cell division in the absence of replication.

Limiting initiation leads to an increased rate of replication fork movement and more frequent replication fork collapse

Replication forks in cells with a fourfold increase in *datA* dosage moved from the origin to the terminus with twice the speed of wild-type replication forks (Fig. 4). Previous work (Torheim *et al.*, 2000) has also shown a faster than wild-type replication rate in a *dnaA(Ts)* mutant (*dnaA204*). From their flow cytometry data of untreated, slowly growing cells with a doubling time of 4 h (Fig. 1A and C in Torheim *et al.*, 2000), we calculated that the C-periods of wild-type and *dnaA204* cells were 80 min and 30 min respectively. Thus, the replication fork rate was more than twofold increased in the *dnaA204* cells. An increased rate of fork movement was also observed in the *dnaA(Cs,Sx)* strain, a mutant isolated as a suppressor of a temperature-sensitive polymerization defect resulting from a mutation in *dnaX* (Gines-Candelaria *et al.*, 1995; Boye *et al.*, 2001). The rate of fork movement in the mutant was twice that in the wild type, at both 34°C and 39°C (data not shown). These results show that not only a limiting amount of free DnaA protein, but also changes in the DnaA protein by mutation can lead to an increased rate of replication fork movement.

Cells with limiting HNS (Atlung and Hansen, 2002) or IHF (von Freiesleben *et al.*, 2000b) activity have also been reported to replicate with shortened C-periods. The C-period in the *hns206* mutant was found to be about half that of wild type at 30°C (Atlung and Hansen, 2002). A common feature of the *hns*, *ihf* and *dnaA(Ts)* mutants and the wild type harbouring MiniR1-*datA* is that the origin concentration is lower than normal. Thus, an explanation for how an increased rate of fork movement might be possible would simply be that there is an increased res-

ervoir of replication components and nucleotides as a result of the increased cell mass. However, it is also reasonable to suppose that, in these cells with a highly increased initiation mass, the replication process is not properly coupled to the rest of the cellular growth processes.

Assembled replication forks often pause, stall and even collapse during their passage from origin to terminus (Bierne and Michel, 1994; Kuzminov, 1995b; Michel *et al.*, 1997; Rothstein *et al.*, 2000). Such interruptions in replication may be caused by single-strand breaks in the template DNA, either pre-existing or arising from abnormal DNA structures or from collision with protein complexes (Brewer, 1988; French, 1992). However, as replication forks seem to be capable of traversing the distance from origin to terminus twice as fast as they do in the normal situation, it is also possible that most replication pausing or delay in wild-type cells is not caused directly by obstacles. It is probable that properly assembled replication forks have a higher order organization of replication elongation that ensures stability of the replication forks. In the cells with fast replication rates reported here, such organization may be lost. If so, a collapse of forks when encountering an obstacle would be expected to be more frequent and would explain why cells with fast replication rates were more dependent on PriA and RecA functions.

Reduced replication fork speed has also been observed in certain chromosome regions during yeast replication (Cha and Kleckner, 2002). These are regions in which the replication fork pauses in order to provide time for the development of chromosomal features behind the fork, such as installation of nucleosomes and development of intersister connections. It is possible that the *Escherichia coli* replication fork also has regulatory mechanisms that monitor proper organization of sister chromosomes as they emerge from the replication factory and that these mechanisms contribute to the relatively slow progression of wild-type replication forks.

It is not clear whether the DnaA protein is actively involved in regulating the rate of fork movement. Increased fork speed seems to occur when the initiation mass is increased, whether DnaA is limiting or not. Data presented here, as well as previously (Skarstad *et al.*, 1989; Atlung and Hansen, 1993), show that replication forks move more slowly upon overproduction of the DnaA protein. However, this effect might not be direct either. It is possible that replication forks formed when DnaA is overproduced, and which form 'too early' relative to other cell cycle events, experience a scarcity of replication components or nucleotides. We attempted to answer whether this was the reason for the slow rate of replication, or whether DnaA could be involved directly at the fork. The rate of replication was measured after overproduction of

DnaA in strain EC::71CW, in which chromosome replication is initiated independently of DnaA from the origin of plasmid R1 (Bernander *et al.*, 1991). As this strain does not overinitiate when DnaA is provided in excess, there should be no scarcity of replication components because of 'too early' replication. However, even though replication fork rate was found to be somewhat decreased (data not shown), the rate of fork movement in strain EC::71CW was extremely slow when DnaA was both overproduced and not (data not shown). Therefore, replication forks in this strain cannot be taken to be representative of wild-type replication forks.

Very low levels of free DnaA protein leads to induction of the SOS response and a requirement for recombination

A dosage of *datA* of about 12-fold led to incomplete run-out of chromosome replication after rifampicin treatment (Figs 1D and 3G). Cells were sensitive to UV light (Fig. 6) and dependent on recombination genes and *priA* for survival (Table 7). These results indicate that a severe lack of DnaA protein causes instability of replication forks. Under these conditions, the SOS response was induced, and a severe growth impairment was observed, supporting the notion that forks are dependent on DnaA for proper function. It must be noted, however, that under these conditions of severe stress, many different responses may be elicited.

Experimental procedures

Bacterial strains and plasmids

All bacteria used were *E. coli* K-12 and are listed in Table 8. All plasmids used have been described previously (Morigen *et al.*, 2001). Transformations were performed by electroporation. Viability of transformants was measured by restreaking the transformants on the same selective plates. Transformation efficiency was determined by formed colony number per µg of DNA. For JJC1398, a *priA2::kan* strain carrying a *PriA*-producing plasmid, competent cells were prepared from a colony cured of the plasmid.

Media

Cells were grown in AB minimal medium (Clark and Maaløe, 1967) supplemented with 10 µg ml⁻¹ thiamine, 0.2% glucose, 0.5% casamino acids (ABTG-CAA). Ampicillin, chloramphenicol, kanamycin and tetracycline were added to final concentrations of 50 µg ml⁻¹, 25 µg ml⁻¹, 25 µg ml⁻¹ and 25 µg ml⁻¹, respectively, when required for selection.

Flow cytometry

Exponentially growing cells (OD₄₅₀ = 0.15) were treated with 150 µg ml⁻¹ rifampicin (Fluka) and 10 µg ml⁻¹ cephalixin (Eli Lilly) for four or five generations to complete ongoing rounds of replication. Rifampicin inhibits transcription, which results in inhibition of initiation of DNA replication, whereas cephalixin inhibits cell division (Skarstad *et al.*, 1986; Boye and Løbner-Olesen, 1991). In the presence of these drugs, cells end up with an integral number of chromosomes (Skarstad *et al.*, 1986), which represents the number of origins at the time of drug treatment. In a culture of cells with synchronous initiation, the integral number of chromosomes is two, four or eight but, with asynchronous initiation, additional peaks representing cells with three, five, six or seven chromosomes appear.

Cells treated with rifampicin and cephalixin or exponentially growing cells were collected and resuspended in TE buffer, then fixed with 77% ethanol at 10× dilution. The cells were washed in 0.1 M phosphate buffer (PB), pH 9.0, and stained overnight (4°C) in 1.5 µg ml⁻¹ fluorescein isothiocyanate (FITC) in the same buffer (Wold *et al.*, 1994). Then the cells were washed in 0.02 M phosphate-buffered saline (PBS), pH 7.5, and resuspended in the same buffer. The DNA was stained in 1.5 µg ml⁻¹ Hoechst 33258. Staining of the DNA standard was performed as described previously (Torheim *et al.*, 2000).

Flow cytometry analysis was then performed using a FAC-Star+ instrument (Becton Dickinson) equipped with an argon ion laser and a krypton laser (both Spectra Physics) as described previously (Torheim *et al.*, 2000). The average cell mass, determined as average FITC fluorescence per cell, was calculated by taking the average of the FITC fluorescence intensity distribution. The average DNA content per cell, determined as average Hoechst fluorescence per cell, was calculated by taking the average of the Hoechst fluorescence intensity distribution. The average DNA per mass was found by dividing average DNA content per cell by average cell mass.

Table 8. Bacterial strains.

Strain	Relevant genotype	Source/reference
MG1655	Wild type	Guyer <i>et al.</i> (1981) and Jensen (1993)
KS9921	MG1655 $\Delta seqA$	Torheim <i>et al.</i> (2000)
ALO1186	MG1655 <i>lacZ::Tn5</i> , <i>sfiA-lac</i>	This work
DM2225	<i>ntl-1</i> , <i>rpsL31</i> , <i>thi-1</i>	Fram <i>et al.</i> (1989)
DE402	DM2225, <i>lexA3</i>	Fram <i>et al.</i> (1989)
JJC1398	AB1157 <i>hsdR</i> , Thr ⁺ , Pro ⁺ , <i>sfiA11</i> , <i>priA2::kan/pAM-priA</i>	Grompone <i>et al.</i> (2003)
JC2924	<i>recA56</i> , <i>his-4</i>	Vadi (1983)
JC5519	<i>recB21</i> , <i>recC22</i> , <i>his-4</i>	Suter and Jaeger (1982)
AB20	C600 <i>dnaA(Cs,Sx)</i> <i>zib::Tn10</i>	Boye <i>et al.</i> (2001)
AB21	C600 <i>zib::Tn10</i>	Boye <i>et al.</i> (2001)
EC::71CW	<i>metB1</i> , <i>nalA</i> , <i>relA1</i> , <i>spoT1</i> λ r, F ⁻ , $\Delta oriC::pOU71$, <i>bla</i> , Rep-Td	Bernander <i>et al.</i> (1991)

Measurement of the replication rate

Samples were taken at different time intervals (0, 20, 40, 60, 90, 120 and 180 min) after the addition of rifampicin ($150 \mu\text{g ml}^{-1}$) and cephalixin ($10 \mu\text{g ml}^{-1}$) to exponentially growing cells ($\text{OD}_{450} = 0.15$). The samples were analysed using flow cytometry through the same procedures mentioned above.

Rifampicin inhibits the initiation of DNA replication, but not the elongation of DNA chain. In the presence of the drug, therefore, the DNA histograms gradually changed from the continuous broad distribution characteristic of a culture in steady-state growth to a distribution containing multiple sharp peaks representing cells with integral numbers of chromosomes. The kinetics of this change reflect the rate of replication fork movement (Skarstad and Wold, 1995).

Treatment of cells with chloramphenicol to view the nucleoid distribution

Exponentially growing cells were treated for 15 min with chloramphenicol ($300 \mu\text{g ml}^{-1}$) to condense the nucleoids (Begg and Donachie, 1991) and cephalixin ($10 \mu\text{g ml}^{-1}$) to inhibit cell division during nucleoid condensation. Cells were fixed in 70% ethanol. Microscope slides were prepared and stained as described previously (von Freiesleben *et al.*, 2000a). Visualization of the cells was performed using a Zeiss Axioplan 2 phase-contrast/fluorescence microscope equipped with a $63\times$ objective and BP365/12 excitation filter. Pictures were taken and stored using a Micromax CCD camera (Princeton Instruments). More than 280 cells for each experiment were included in the nucleoid counting.

Immunoblotting

Exponentially growing cells ($\text{OD}_{450} = 0.20$) were harvested by centrifugation at 4°C and resuspended in Tris-EDTA buffer containing 1% SDS and glycerol and boiled for 5 min. Total protein concentration was determined by a colorimetric assay (BCA kit; Pierce). A buffer containing SDS and β -mercaptoethanol was added. Preparation of DnaA standards was the same as mentioned previously (Torheim *et al.*, 2000). Fixed amounts of cell extracts and DnaA standard were subjected to SDS-PAGE (12%). The protein was transferred to a polyvinylidene difluoride (PVDF) membrane by semi-dry blotting. The membrane was probed with DnaA anti-rabbit antibody using standard procedures. Detection was performed with a fluorescence kit (ECF kit; Amersham Pharmacia Biotech). The membrane was scanned on a Storm 840 (Molecular Dynamics), and quantification was carried out using IMAGEQUANT software (Molecular Dynamics). A DnaA standard curve was made, and the concentrations of DnaA in the cell extracts were determined by comparison with the standard curve.

Southern hybridization

Southern hybridization was performed as described previously (Løbner-Olesen and von Freiesleben, 1996) to measure the gene dosage of *oriC*, *datA* and *terC*. Preparation of total cellular DNA and the labelled probes *oriC*, *terC* and *bla*

for *datA* plasmids, as well as visualization and quantification of radioactive bands, was performed as mentioned previously (Morigen *et al.*, 2001).

UV treatment

Transformed cells were plated on selective plates. Then, the plated cells were immediately irradiated by UV light (254 nm) at the following doses: 0, 15, 30, 45, 60, 75 and 90 J m^{-2} . The UV-irradiated cells were incubated in the dark. For induction of the SOS response (Table 6), exponentially growing cells ($\text{OD}_{450} = 0.10$) in ABTG-CAA medium were irradiated by UV light (254) for 40 J m^{-2} directly in an open plate.

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